Assignment 1: Metamorphic Testing of a Multi-Agent Housing-Trends Application

Final report

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1 Abstract

We built a multi-agent application that reports twelve-month price trends for residential listings in Dallas, Texas, on a frozen, reproducible snapshot, and tested it with ten metamorphic relations (MRs) from all three families (invariance, monotonic, directional). Each MR was run on 30 distinct base inputs under three fixed seeds, and violation rates are reported with 95% Wilson intervals. With the rule-based (offline) stand-in for every LLM node, four MRs never fail (upper bound 11.4%). The three most-violated relations (MR2 duplication, 100%; MR4 remove the most expensive listing, 53.3%; MR7 double one ZIP, 43.3%) were diagnosed by switching the pipeline's adaptive loops off and on. All their violations disappear with the loops off. Pooling alone reproduces every one of them, and a replay exposed a second, unanticipated fault in the anomaly Diagnoser: its evidence is chosen by listing id under a fixed cap. Two of the top three are genuine aggregation faults; MR2 is mostly an ill-conceived relation plus that Diagnoser fault. Re-running all ten MRs with Qwen 2.5 7B in every LLM node (900 trials, 0 errors) gives higher rates for MR4 (83.3%) and MR7 (70.0%), consistent with the model borrowing neighbouring ZIPs far more often than the rule-based policy does. It also exposes a failure the offline run cannot show: MR10 (scale every price by k), which holds 0/90 offline, is violated on 73.3% of base inputs with Qwen. An ablation and a three-trial control indicate that the cause is the LLM's neighbour-ZIP ranking in Pooling, which can change with the prompt text, even for a 10^{-6} relative price change. With Pooling off, MR10 holds 90/90.

2 Problem and output contract

The application answers: *how have prices moved over the past twelve months for listings matching these criteria?* Criteria are a set of Dallas ZIP codes, an optional price band, minimum beds and baths, optional size and property-type filters, and a twelve-month window inside the snapshot's coverage.

Output contract. Every MR is a statement about these numbers and nothing else (`mr/contract.py`):

1. **Overall series**: one row per month of the window, over all requested ZIPs, own listings only (pooled listings are never double-counted): `price` (median list price), `ppsf` (median price per square foot; listings without a usable size are skipped), `n` (active listings behind the month, after exclusions), and `mom_price_pct` (month-over-month % change of `price`).
2. **Per ZIP-month table**: one row per requested ZIP × month: `price` and `ppsf` (the bucket statistic, including pooled neighbours when Pooling widened a thin bucket), `n_own`, `n_used`, and `mom_price_pct` within the ZIP.

Notes, loop counters and the decision trace are diagnostics, not part of the contract.

3 System architecture

3.1 Agents

The pipeline is orchestrated with LangGraph (`housing_trends/graph.py`). The design rule throughout is that an LLM supplies *judgement* and code does all *arithmetic*. Every LLM reply is treated as untrusted: it is validated against the options code offers, the model gets the errors back once, and otherwise the node falls back to a deterministic rule-based policy (the *stand-in*, used for the offline runs).

Agent	LLM decision	Responsibility
Input Handler / Parser	Free text only	Form input goes straight to a deterministic validator. Free text goes to the Parser Agent, whose JSON goes through the *same* validator (one repair attempt).
Scanner	Neighbour ranking	Retrieves listings per ZIP-month bucket and computes, exactly, what each widening move would add (`thin`, `reachable`, `exhaustive`). The LLM ranks the most comparable neighbouring ZIPs to pool from.
Statistician	None	Computes median / trimmed mean / median-of-medians per bucket and overall, plus month-over-month %. It never judges its own output.
Critic	Stop / iterate	Owns every routing decision: coverage → Pooling, price jump → robust recompute, persistent jump → Diagnoser, else render. Screens listings for outliers (robust z of log price > 3.5).
Pooling	Move choice	For thin buckets, chooses per bucket class among legal moves: pad months, borrow ZIPs, relax a non-protected filter.
Anomaly Diagnoser	Verdicts	Judges outlier listings as `bad_listing` (excluded), `legit_rural` (kept; ZIP uses the trimmed mean) or `unexplained` (excluded and flagged), and explains each anomaly.
Renderer	None	Presentation only (Streamlit dashboard, `app_input.py`): trend, coverage, warnings, decision log.

Table 1: Agents as implemented. Thresholds (`housing_trends/state.py`): `MIN_LISTINGS`= 20, `JUMP_PCT`= 30%, trimmed-mean fraction 0.1, at most 3 rescans, 1 recompute, 2 Diagnoser rounds.

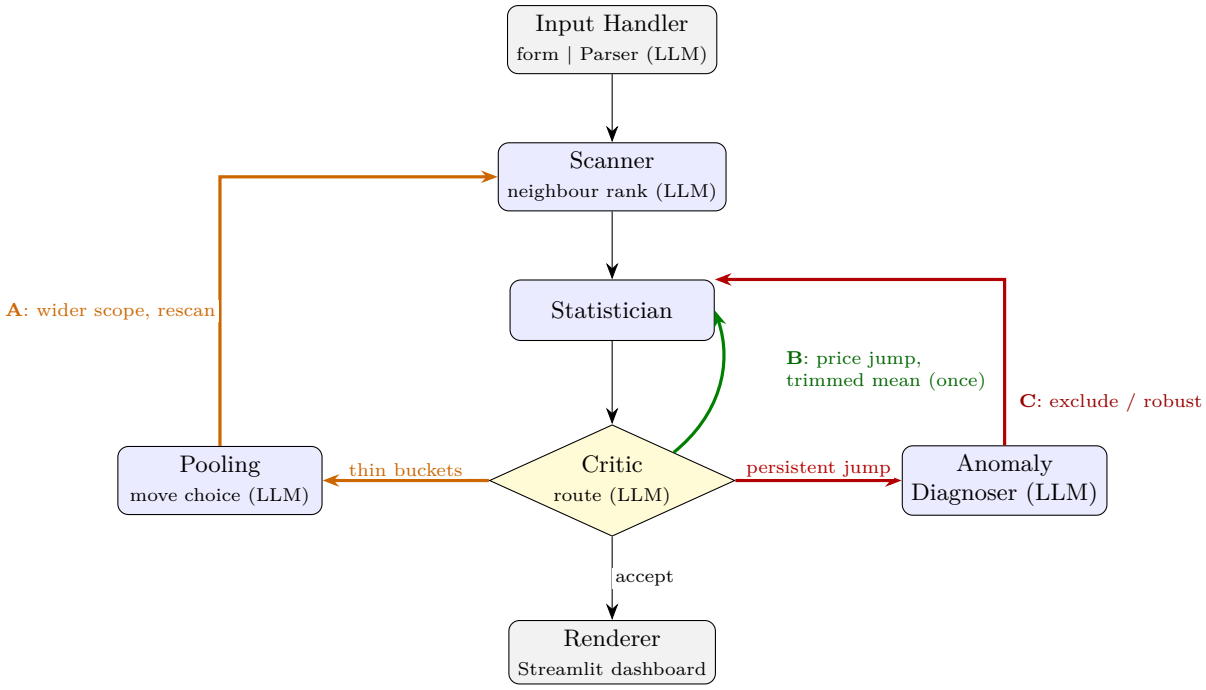


Figure 1: Implemented control flow (`housing_trends/graph.py`). Coloured edges are the three cycles, each bounded by a counter: **A** coverage (Critic → Pooling → Scanner, ≤ 3 rescans), **B** method (Critic → Statistician, robust recompute ≤ 1), **C** outlier (Critic → Diagnoser → Statistician, ≤ 2 rounds). The Critic takes the first applicable step on each visit, so the loops are ordered: coverage, then trend, then outliers.

3.2 Control-flow graph

Figure 1 shows the implemented graph with its three cycles. Figure 4 reproduces the diagram from our design note for comparison (§3.6).

3.3 Why the loops are necessary, not decorative

- **Cycle A (coverage).** At the 20-listing threshold, 304 of 504 ZIP-month buckets are thin (60%) on the default scope. Median noise is about ±13% with 5 listings and ±6.5% with 20 (ZIP 75241). Thinness is only known after a scan, and how much widening a bucket needs (229 buckets need ±1 month, 44 need ±2, 5 need ±3, 26 also need ZIPs) is only known after the rescan. So the loop sets a per-bucket depth and verifies it, rather than guessing it up front. We concede that the loop is partly decorative in one respect: time padding alone fixes 278 of 304 thin buckets, so the *choice* of move is rarely hard (NOTEBOOK,

2026-10-08; AI_USE case 8).

- **Cycle B (method).** A > 30% price jump is visible only after the statistic is computed. 53 of 451 ZIP-months jump. The trimmed mean removes 8 of them (45 remain), and pooling ±3 months removes most. The recompute tells the system whether the jump was an artefact of the plain median.
- **Cycle C (outliers).** Medians hide typos: removing all 71 injected typo listings changes only 33 of 504 bucket medians. A jump that survives the robust recompute therefore needs listing-level evidence, and the Diagnoser's verdict changes *what the Statistician does next* (exclude a listing, or switch the ZIP to the trimmed mean). This is the assignment's "genuine dependency" (c).

3.4 A loop firing at run time

The trace below (`results/traces/short_example.txt`, 2 ZIPs, stand-in LLM) fires all three cycles in one run: one rescan, one robust recompute and one Diagnoser round.

```
critic: coverage: 3/24 buckets below 20 listings (3 fixable) -> Data Issue, route to Pooling (rescan 1/3)
pooling: rescan 1: 3 fixable of 3 thin: time +1 month for 3 bucket(s)
scanner: 24 buckets, 3 widened, 0 thin (<min), ... depth used: time_pad max 1, zip_extra max 0
critic: coverage OK (0 thin of 24; 1 rescan(s) used)
critic: trend: 6 price jump(s) > 30% in 2 ZIP(s) -> robust recompute (trimmed mean 0.1) ... (recompute 1/1)
critic: trend: robust recompute resolved 2 of 6 jump(s); 4 persist
critic: trend: 4 persistent anomaly(ies) -> Diagnoser round 1/2 (4 sent, 2 outlier suspect(s))
diagnoser: listing 13442 (75219, 2025-12, $5,776,500) -> bad_listing: price is 13.54x the ZIP median
diagnoser: listing 14698 (75225, 2026-07, $9,676,300) -> bad_listing: price is 15.68x the ZIP median
statistician: method=median, 24 buckets (0 empty), 0 override(s), 2 excluded id(s)
critic: trend: 6 jump(s) persist after 1 Diagnoser round(s) (no new outlier suspects ...) -> accept and flag as unresolved
```

3.5 User interface

The dashboard (`app_input.py`, `make app`) runs on the frozen snapshot. Figure 2 shows the criteria form and the result view: the twelve-month trend, coverage metrics and warnings, and the agent decision log. Figure 3 shows the trend charts themselves.

3.6 Changes from the design note

Table 2 lists every difference between the design note and the system we tested. Each was checked against the code. Figure 4 shows the original diagram.

Table 2: Design note versus implementation.

Aspect	Design note	Implementation, and why
Loops	Two cycles: A (Pooling) and B (one recompute). The Diagnoser ends the flow ("Explained?" → Render or Flag).	**Three cycles.** New Cycle C: the Diagnoser's verdicts go *back* to the Statistician (exclude, or trimmed mean), at most 2 rounds. Without it the Diagnoser could only annotate, never correct. "Flag" became the report's unresolved list.
Rolling-window node	Separate node between Critic and Statistician ("only the first time").	Folded into the Critic's trend check. "Only the first time" is the counter `MAX_RECOMPUTES`= 1, and the recompute is a specific method: trimmed mean 0.1 over the whole series of each flagged ZIP.
What counts as a jump	Big jumps in median price *or* active-listing count.	**Price only.** Our snapshot's volumes are too volatile: 74% of healthy ZIP-months change count by > 30% (NOTEBOOK 2026-10-09). Count changes are reported, not flagged.
Diagnoser input	Composition shifts, property mix, sample size; output is explanatory text.	Listing-level evidence: the Critic's outlier screen (robust z of log price > 3.5, at most 8 suspects) plus ZIP percentiles of lot, house size and price. The output is verdicts (which change the computation) *and* one explanation per anomaly.

Aspect	Design note	Implementation, and why (continued)
Pooling strategy	Fixed rules: small sample → time; concentrated → ZIPs; restrictive filter → relax; exhaustive → stop.	Code computes the legal moves per thin bucket (cap not reached and the move adds a listing). The LLM picks one move per bucket *class*, validated, with the rule order as fallback. Depth is per bucket, and **healthy buckets are never widened** (a first version altered 98% of them). Caps: 6 ZIPs, ±3 months, 3 rescans; price, beds, baths protected.
Neighbour ZIPs	"Adjacent ZIPs".	Geographic ordering from frozen centroids is implemented (`housing_trends/geo.py`), but the centroid file is not in the snapshot, so all reported runs use the **numerically nearest** ZIPs (threat, §8).
Data	Kaggle USA Real Estate, filtered to DFW (75xxx), with listing dates; ZHVI for monthly values.	Kaggle has *no listing date*, so a real 12-month trend was impossible. We built a **calibrated synthetic snapshot** of Dallas city only, with 24 months of coverage (§4).

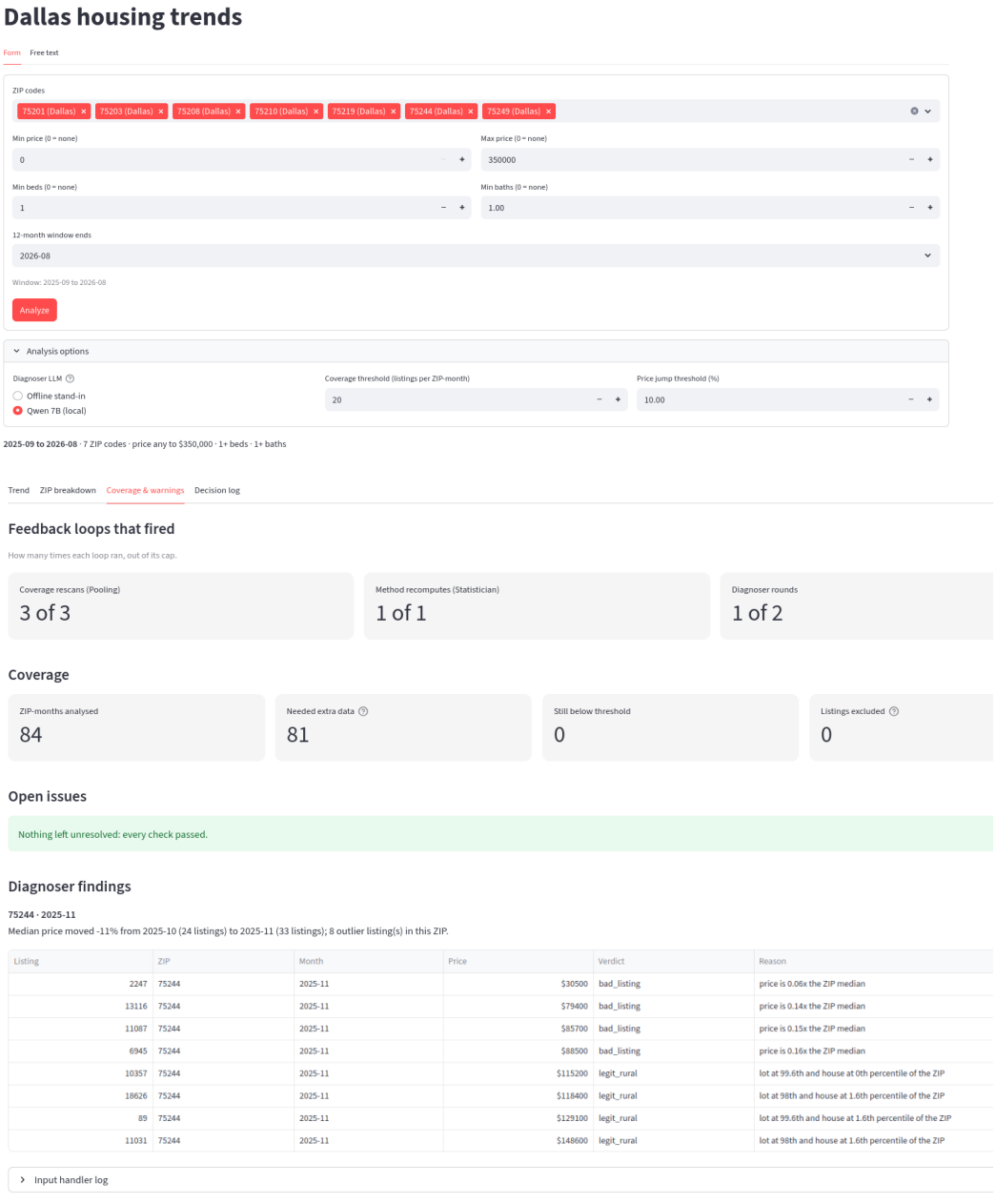


Figure 2: Streamlit dashboard. Top: criteria form (ZIPs, price band, beds, window). Bottom: trend, coverage metrics, warnings and decision log for the same query.



Figure 3: Streamlit dashboard, trend view: median list price and median price per square foot by month for the same query.

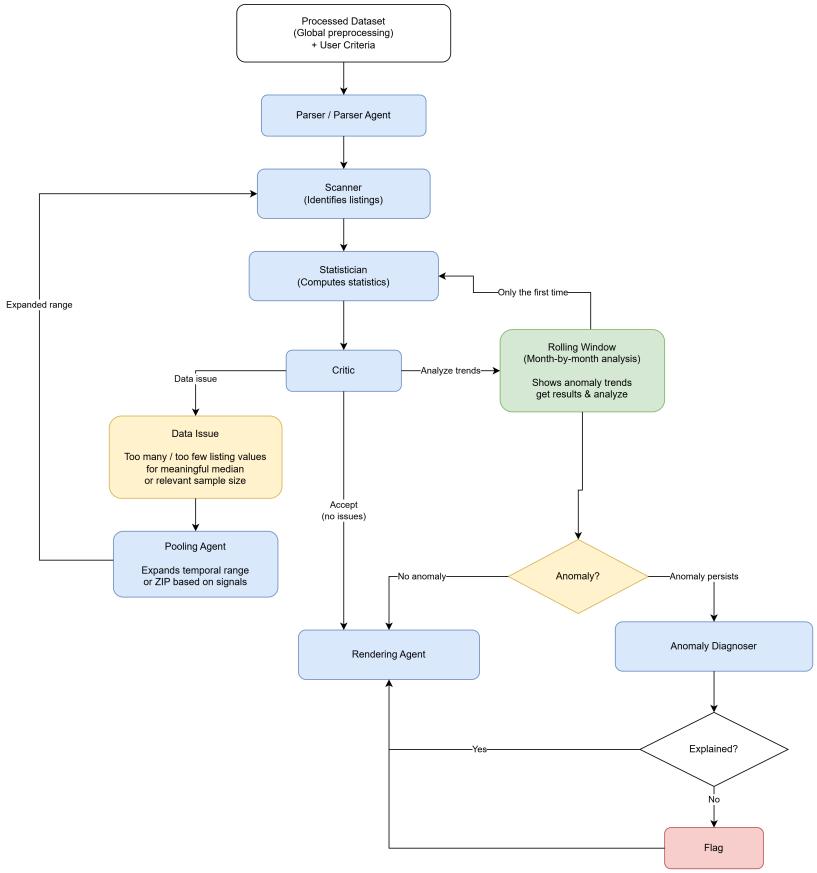


Figure 4: Architecture diagram from our design note (`TSE_agentic_architecture_Bavya_Arushi.pdf`, Fig. 1), reproduced for comparison with Figure 1.

4 Data source and frozen snapshot

- **Sources.** Kaggle USA Real Estate (realtor.com scrape, `realtor-data.zip.csv`, about 2.2M rows; `https://www.kaggle.com/datasets/ahmedshahriarsakib/usa-real-estate-dataset`) and Zillow ZHVI (city level, middle tier, smoothed, seasonally adjusted, through 2026-08; `https://www.zillow.com/research/data/`).
- **Problem.** Kaggle has no listing date. `prev_sold_date` on sold rows falls entirely in 2021-10-18 to 2022-05-02, which leaves about six usable months, so a twelve-month trend cannot be built from it directly.
- **Solution** `python scripts/build_snapshot.py --cities Dallas --window 24`, seed 20261016. price $= f_{\text{zip}} \times \text{ZHVI}(\text{city}, \text{month}) \times$ size adjustment $\times$ noise. ZHVI supplies the month-to-month movement, and Kaggle supplies ZIP price levels, size elasticity (0.8177), noise ($\sigma = 0.1695$), property mix and volume, calibrated on 26,838 real rows.
- **Snapshot.** 19,976 listings, 42 ZIPs, window 2024-09 to 2026-08. Injected faults with ground-truth labels that no agent reads (`listings_dfw_truth.csv`): 29 price typos $\times 10$, 42 typos $\div 10$, and 60 legitimate rural low-value listings.
- **Reproducibility.** SHA-256 hashes are recorded in `data/snapshot/snapshot_manifest.json`. A rerun and a run on another machine produced byte-identical files.

5 Metamorphic relation catalog

Tables 3 and 4 summarise the ten relations: Table 3 gives the transformation, relation, subdomain and tolerance, and Table 4 gives the justification (why each relation should hold) and the fault it would expose. The full catalog is generated from the code that executes them (`docs/mr_catalog.md`, from `mr/relations.py`), so it cannot drift. An MR is a hypothesis: a violation is a symptom with a cause to find, not automatically a bug. Two relations from an earlier twelve-MR catalog (reorder listings; add a fake month outside the window) were withdrawn because they were redundant. Random streams are keyed per relation, so the remaining relations perturb exactly the same months, ZIPs and listings as before the renumbering.

Table 3: MR catalog: transformation, relation on the output contract, subdomain and tolerance.

MR	Family	Transformation	Relation (on the contract)	Subdomain	Tolerance
MR1	Invariance	Sort listings by price inside every ZIP-month (adversarial order)	Whole contract unchanged	All	Relative 10^{-9}
MR2	Invariance	Each ZIP's listings $\times N_{\text{zip}}$, $N \in [1, 10]$, fresh ids	Per ZIP-month price, ppsf, mom % unchanged; `n_own` exactly $N_{\text{zip}}\times$	All (expected to fail where buckets cross the threshold)	Relative 10^{-9}; exact count
MR3	Monotonic	Add \$50,000 to every price in one month	That month's median rises by about \$50,000; no month's median falls	No price band	Within 1% of X; no fall $> 10^{-6}$
MR4	Monotonic	Remove the most expensive in-scope listing of each ZIP	No ZIP-month median increases	All	No rise $> 10^{-6}$
MR5	Monotonic	Widen the price band by 25%	No month's count falls	Has a band	Exact
MR6	Monotonic	Keep a strict subset of the ZIPs	No month's count rises	≥ 2 ZIPs	Exact
MR7	Directional	Double all prices in one ZIP	That ZIP's price and ppsf double; its counts and mom % and every other ZIP's rows unchanged	No price band	Relative 10^{-9}
MR8	Directional	Price of one counted listing $\times 10$	No overall or ZIP-month median moves $\geq 15\%$; if the Diagnoser examines the ZIP, the listing is a suspect	No price band	$< 15\%$; membership
MR9	Directional	Replace a month by a copy of the previous month	The two months have equal price, ppsf, count; mom % $= 0$	All	Relative 10^{-9}

MR	Family	Transformation	Relation (on the contract)	Subdomain	Tolerance
MR10	Directional	Scale all prices by $k \in \{1.5, 2, 3\}$	Prices and ppsf scale by k; counts and mom % unchanged	No price band	Relative 10^{-9}

Table 4: MR catalog: why each relation should hold (justification) and the fault it would expose if violated.

MR	Why it should hold	Fault exposed if violated
MR1	Statistics depend on the set of listings, not their order.	Position-dependent logic (head/tail slicing, id or position tie-breaks) that a random shuffle rarely triggers.
MR2	Replicating a sample leaves its median unchanged; only counts scale.	Estimators that are not scale-free in sample size; coverage logic that reacts to the count; id-keyed exclusions that do not follow a listing's copies.
MR3	Shifting every value by X shifts the median by exactly X.	The robust recompute reacting to the injected shift (median replaced by trimmed mean); month bucketing errors.
MR4	Removing the maximum of a sample cannot raise its median.	Buckets pushed across the coverage threshold (Pooling borrows more expensive neighbours); any estimator not monotone in its sample.
MR5	A superset of listings cannot have fewer members.	Wrong inclusive/exclusive bounds; exclusions that remove more listings after widening.
MR6	A subset of ZIPs matches a subset of listings.	ZIP scoping errors; pooled listings leaking into the own count.
MR7	A ZIP's own statistics scale with its prices, and other ZIPs' data are untouched.	Cross-ZIP leakage: Pooling borrows a changed ZIP's listings into its thin neighbours.
MR8	One outlier should not move a robust median by 15%.	A fragile estimator, or a single listing flipping a ZIP to the trimmed mean; an outlier screen that misses a 10× listing.
MR9	Identical data should give identical statistics.	Identical months treated differently (pooling or recompute depending on neighbouring months; id-keyed exclusions).
MR10	Relative quantities are scale-free, so levels scale by k and percentages do not change.	Absolute (non-relative) thresholds or rounding in the Critic, Diagnoser evidence or Statistician.

6 Statistical protocol and results

Protocol. A reproducible pool of 240 distinct criteria sets (ZIP subsets × price band × beds/baths × window), each matching at least 50 listings, is drawn with pool seed 20261016 (`mr/inputs.py`). Each MR takes the first 30 pool members in its subdomain. Each base input is run under seeds 101, 202 and 303. A seed drives the transform's random parts (shuffle, ids, which ZIP or month, N), and with the real LLM it is also the model's seed, shared by the source and follow-up runs. A base input counts as *violated* if any of its seeds violates the relation. The headline rate is violated base inputs / base inputs, with a 95% Wilson score interval (`mr/stats.py`), which behaves well near 0 and 1 where most of our rates lie. The per-run rate is also reported. Crashes are recorded as errors, and follow-ups that match no listing as skipped. Neither counts as a pass or a violation.

When is it enough? With 30 base inputs, an MR with no violations has a Wilson upper bound of 11.4%. That is tight enough to conclude the relation holds at least 88% of the time on this subdomain. For the top three the intervals lie far from 0 (lower bounds 27–89%), so 30 base inputs settle that each violation is systematic.

Real LLM (Qwen 2.5 7B). To measure the system with a non-deterministic model, we re-ran all ten MRs with every LLM node served by Qwen 2.5 7B Instruct (Q4_K_M, Ollama, temperature 0, model seed = trial seed), with the same 30 base inputs and 3 seeds (900 trials; about 34 LLM calls per trial on one GTX 1080 Ti). The run was done in three parts (`results/mr_llm/` for MR2, MR4, MR7; `results/mr_llm_rest/` for the other seven, with MR6 and MR10 completed in `results/mr_llm_rest2/` after the first attempt was interrupted). Every reply is cached (`results/llm_cache/`), so the runs replay without a GPU. For MR2, MR4 and MR7 we confirmed this: replaying from the committed cache reproduced the violation counts exactly, with no live model call and 0 failed calls.

With the model in the loop, MR3, MR4, MR7, MR8 and MR10 fail *more* often than with the rule-based stand-in (MR4 runs 75 vs 48, MR7 61 vs 24, MR10 64 vs 0). The ranking by rate becomes MR2, MR4, MR10, MR7, MR3. MR10 and MR7 differ by one base input (22 vs 21 of 30) with almost identical intervals, so their order is not meaningful. MR7 stays in the headline top three because its cause is structural and reproduces offline, whereas MR10's failure exists only with the LLM in the loop (§7). The Qwen and offline intervals for

MR	Family	Bases	Violating bases	Rate (95% Wilson CI)	Violating runs	Skipped
MR1	Invariance	30	0	0.0% (0.0–11.4)	0 / 90	0
MR2	Invariance	30	30	100.0% (88.6–100.0)	90 / 90	0
MR3	Monotonic	30	5	16.7% (7.3–33.6)	7 / 90	0
MR4	Monotonic	30	16	53.3% (36.1–69.8)	48 / 90	0
MR5	Monotonic	30	0	0.0% (0.0–11.4)	0 / 90	0
MR6	Monotonic	30	0	0.0% (0.0–11.4)	0 / 89	1
MR7	Directional	30	13	43.3% (27.4–60.8)	24 / 90	0
MR8	Directional	30	2	6.7% (1.8–21.3)	3 / 90	0
MR9	Directional	30	3	10.0% (3.5–25.6)	6 / 90	0
MR10	Directional	30	0	0.0% (0.0–11.4)	0 / 90	0

Table 5: Headline results, offline stand-in for every LLM node (`results/mr/summary.md`; reproduce with `make mr`). Seeds 101, 202, 303; pool seed 20261016; snapshot SHA-256 `2aa4f1fd...`; 0 errors in 900 trials.

	Qwen 2.5 7B		Offline stand-in (Table 5)	
MR	Violating bases, rate (95% CI)	Runs	Violating bases, rate (95% CI)	Runs
MR1	0/30, 0.0% (0.0–11.4)	0/90	0/30, 0.0% (0.0–11.4)	0/90
MR2	30/30, 100.0% (88.6–100.0)	90/90	30/30, 100.0% (88.6–100.0)	90/90
MR3	19/30, 63.3% (45.5–78.1)	40/90	5/30, 16.7% (7.3–33.6)	7/90
MR4	25/30, 83.3% (66.4–92.7)	75/90	16/30, 53.3% (36.1–69.8)	48/90
MR5	0/30, 0.0% (0.0–11.4)	0/90	0/30, 0.0% (0.0–11.4)	0/90
MR6	1/30, 3.3% (0.6–16.7)	1/89	0/30, 0.0% (0.0–11.4)	0/89
MR7	21/30, 70.0% (52.1–83.3)	61/90	13/30, 43.3% (27.4–60.8)	24/90
MR8	9/30, 30.0% (16.7–47.9)	13/90	2/30, 6.7% (1.8–21.3)	3/90
MR9	4/30, 13.3% (5.3–29.7)	6/90	3/30, 10.0% (3.5–25.6)	6/90
MR10	22/30, 73.3% (55.6–85.8)	64/90	0/30, 0.0% (0.0–11.4)	0/90

Table 6: All ten MRs with every LLM node served by Qwen (`results/mr_llm/`, `results/mr_llm_rest/`, `results/mr_llm_rest2/`). Same 30 base inputs and seeds 101, 202, 303; 900 trials, about 30,300 LLM calls (the exact count depends on how trials are scheduled across workers), 0 failed calls, 0 errors, 1 skipped (MR6).

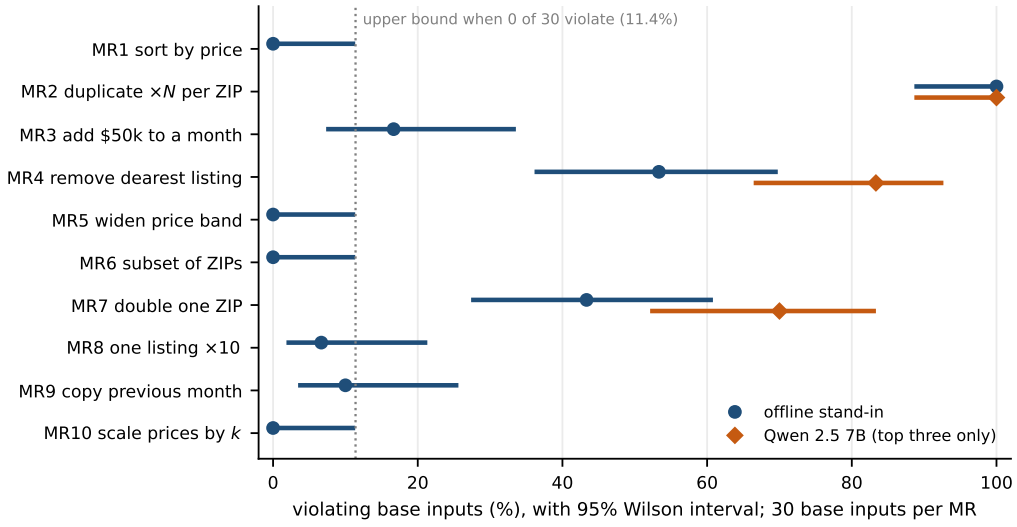


Figure 5: Violation rate per MR with 95% Wilson intervals (data of Tables 5 and 6). The dotted line marks the 11.4% upper bound that 0 violations in 30 base inputs can support.

MR4 barely overlap (66.4 vs 69.8), and for MR7 they overlap more (52.1 vs 60.8). We read this as consistent with, not proof of, a real difference, because overlapping Wilson intervals are not a significance test. The likely cause is visible in the cached replies. Of the 2,617 Pooling move choices (one per bucket class) in the original Qwen run (MR2, MR4, MR7), 957 (36.6%) borrow neighbouring ZIPs. The stand-in pads months whenever that suffices, which on our data is 278 of 304 thin buckets. Borrowing ZIPs is exactly the move that leaks other markets into a ZIP's median (§7). In that run the validator rejected 12 invalid Qwen moves (`"none"` or empty) and fell back to the rule.

Non-determinism across seeds. In 272 of the 300 (MR, base input) pairs, all three seeds agreed: each base input either held under all seeds or failed under all seeds. The 28 mixed pairs sit in the MRs whose transform draws differ by seed: MR3 (12), MR8 (9), MR9 (4), and one each in MR6, MR7 and MR10. At temperature 0, changing Qwen's seed barely changed its decisions, so the repetition measured transform variation, not model non-determinism (§8).

7 Root-cause analysis of the three most-violated MRs

Method. Two independent lines of evidence, both on the same 30 base inputs and 3 seeds (`scripts/diagnose_top3.py`, outputs in `results/mr_diagnosis/`; single trials replay with `scripts/mr_replay.py`): (1) **ablation**: `min_listings`= 0 disables Cycle A, and `jump_pct`= 10^9 disables Cycles B and C; (2) **bucket attribution**: for every flagged ZIP-month in 59 replayed violating trials, whether it was pooled, recomputed with the trimmed mean, or had exclusions.

	Violating runs (of 90)				Violating ZIP-months			
MR	Full	Loops off	Pooling off	Jump loops off	Total	Pooled	Trimmed	Plain median
MR2	90	**0**	18	90	150	133	58	**0**
MR4	48	**0**	36	48	46	46	17	**0**
MR7	24	**0**	0	24	65	65	25	**0**

Table 7: Ablation (left) and bucket attribution (right) for the top three MRs, offline stand-in.

What this rules out. With all loops off, every relation holds on 270 of 270 trials, with the same transforms, checks, tolerances and listings. So the cause is neither a *harness bug* nor a *data artifact*: the plain median satisfies every relation on these listings, and not one violating bucket is a plain own-listing median. What remains is the adaptive control flow, and the question for each MR is whether it is a genuine fault or an ill-conceived relation.

MR4 (remove the most expensive listing): genuine aggregation fault in the coverage loop

The property is sound: removing the maximum of a sample can never raise its median. Yet all 46 violating buckets are pooled, and Pooling alone reproduces all 48 violating runs. In the worst case (base 3, seed 101, `75216@2025-06`) the source bucket has 0 own and 12 pooled listings (median $415,550). After the removal it borrows 28 neighbour listings and reports **$804,150 (+93.5%)**. In another (base 1, `75203@2025-10`) the median goes from $369,706 to $714,674. *Mechanism*: deleting one listing moves buckets across `MIN_LISTINGS`, so Pooling widens them differently, and the reported "ZIP-month median" becomes the median of a different, more expensive population. Several such buckets have 0 or 2 own listings. They describe neighbours, not the ZIP. This is a fault: the estimator becomes non-monotone, and the published population depends discontinuously on a single listing. *Fix*: keep an own-listing statistic in the contract and report the pooled estimate separately, or publish buckets with fewer than k own listings as low-confidence.

MR7 (double one ZIP): genuine fault, cross-ZIP leakage through Pooling

The cleanest ablation: **Pooling off gives 0 / 90 violated runs**, and Pooling on with the jump loops off gives exactly the full pipeline's 24. All 65 violating buckets are pooled on both sides. For example, `75210@2025-08` (2 own, 20 pooled) goes from $299,050 to $620,407, and `75219@2025-03` (3 own, 18 pooled) from $807,900 to $1,665,214. In 25 of the 65 buckets the follow-up also switched to the trimmed mean: the doubled neighbour created a $> 30\%$ jump that fired Cycle B. So the leakage changes even *which estimator* a neighbour receives. *Contrast*: offline, MR10 (scale every price) holds on 0 / 90 runs, because a global scale keeps pooled populations

proportional. A local change breaks only through borrowing, which pins the cause on Pooling, not on a scale-sensitive threshold. The catalog predicted exactly this contrast. This contrast holds only for the deterministic stand-in: with Qwen, MR10 also fails (§7), so under the model it cannot serve as the control for this argument.

MR2 (duplicate each ZIP N times): mostly an ill-conceived MR, plus a genuine Diagnoser fault

Part 1, ill-conceived (133 of 150 violating buckets are pooled). Replicating a sample leaves its median and fractional trimmed mean unchanged, but the coverage threshold is defined on *counts*. A bucket thin in the source is healthy after $\times N$ and is no longer pooled (41 buckets pooled only in the source; 88 at different depths). For example, base 26, seed 101, `75201@2025-07` goes from 5 own + 17 pooled ($400,050) to 20 own + 0 pooled ($948,500). The relation must be restricted to buckets that are non-thin on both sides, or the threshold stated per unique listing. The catalog anticipated this partial failure.

Part 2, genuine fault: Diagnoser evidence is not duplicate-aware. With Pooling off, MR2 still fails on 18 / 90 runs (10 / 30 bases), and every one of those follow-ups ran a Diagnoser round. Replay of base 0, seed 101 (`min_listings`= 0; ZIP 75232 copied $N = 9$ times):

- *Source*: the Critic's suspect list (cap 8) for 75232 holds the $3,818,800 listing 4795 (13.6× the ZIP median, `bad_listing`) *and* listing 14998 ($95,800, `legit_rural`). Because the ZIP has a legitimate low-value listing, it keeps the trimmed mean (`diagnoser.py:153`). January: **$276,242**.
- *Follow-up*: the nine copies of the $3.8M listing share the largest $|z|$, so **eight copies fill all eight suspect slots**. The `legit_rural` listing is never judged, the ZIP has only `bad_listing` verdicts, and it reverts to the plain median (`diagnoser.py:154-157`; overrides fall 168 → 156 → 144). January: **$267,700**, and every month of 75232 changes. The ninth copy is never excluded.

The data distribution is identical, so the relation is right. The output changes only because evidence is chosen by listing id under a fixed cap. *Fix*: de-duplicate suspects by (ZIP, month, price, size) before the cap, or decide a ZIP's estimator from the share of listings judged legitimate. The 36 MR4 runs that fail with Pooling off also pass through these loops. We expect a similar mechanism (removing a ZIP's maximum changes which listings are suspects), but we have *not* confirmed it with a replay.

MR10 (scale every price) with Qwen: unstable neighbour ranking in Pooling

MR10 is a control by design: every statistic is relative, so scaling all prices by k should scale every bucket by k. Offline it holds on 90/90 runs. With Qwen it is violated on 22/30 base inputs (64/90 runs). The errors are much smaller than the 93–107% jumps of MR4 and MR7 (for example 75246 in 2025-09: expected $1.598M, got $1.743M, a 9% difference). The failure is all-or-nothing per base input: 21 base inputs fail under all three seeds, 8 never fail, and one is mixed.

Ablation (Qwen, 30 base inputs, 3 seeds).

Configuration	Violating runs	LLM calls
Full pipeline	64/90	–
Jump loops off (Diagnoser cannot run)	59/90	2,683
Jump loops off and Pooling off (`min_listings`= 0)	**0/90**	0

Our first hypothesis, that the Diagnoser's verdicts were responsible, was wrong: switching its loops off barely changes the rate. Pooling is necessary: with it off, no LLM call is made and MR10 holds 90/90.

Mechanism. For a thin ZIP the Scanner asks Qwen to rank neighbouring ZIPs to pool, and the Critic and Pooling nodes also choose moves. Replaying base 1, seed 101 (loops off), the source and the ×3 follow-up receive different rankings for the same ZIPs: 75211, 75214 and 75220 get differently ordered pools, and 75223 is pooled only in the follow-up. Different listings enter the medians, so the output moves by a few percent, far above the 10^{-9} tolerance. The rule-based stand-in ranks neighbours by a fixed rule (the numerically nearest ZIPs) and is scale-invariant, which is why it never shows this.

Control: is it the scale or the prompt text? We re-ran MR10 with $k = 1.000001$ (the prices change by 10^{-4}%, the meaning not at all, but the prompt text does), on three base inputs (0, 1 and 5) with seed 101 (`results/mr_llm_ctrl_k1e-6/`). Base inputs 0 and 5, which fail at $k = 3$ and $k = 2$, hold; base input 1 still violates (April 2026: 139 listings expected, 140 returned). So a negligible prompt change alone can change Qwen's ranking, and larger real scale changes make it worse. Three trials support only this qualitative reading; they do not estimate how much of the 73.3% is noise.

Verdict. Not a deterministic logic fault in the pipeline's code: the same Pooling code is correct under a stable ranking. It is a robustness fault of the design: the system lets an LLM choose which neighbours to borrow, and that choice is not a stable function of the data. A fix is to make the ranking rule-based (or cache or canonicalise it per ZIP, independent of the price figures in the prompt), and to rerun MR10 as the check.

MR	Verdict	Decisive evidence
MR4	Genuine aggregation fault (Cycle A)	46/46 violating buckets pooled; 0/90 with loops off; median +93.5% after removing the maximum
MR7	Genuine fault: cross-ZIP leakage (Cycle A)	0/90 with Pooling off; 65/65 pooled; MR10 holds 0/90 offline
MR10 (Qwen only)	LLM instability in Pooling's neighbour ranking (Cycle A)	0/90 offline; 64/90 with Qwen; still 59/90 with the jump loops off; 0/90 with Pooling off
MR2	Ill-conceived on thin buckets *and* genuine Diagnoser fault (Cycle C)	133/150 pooled; with Pooling off 18/90 remain, all after a Diagnoser round; replay shows copies crowding out the `legit_rural` listing

Table 8: Root-cause summary.

8 Threats to validity

Internal validity.

- The headline table uses the deterministic stand-in. Its three seeds vary only the transforms, not the agents, so it under-samples the non-determinism the assignment targets. The Qwen run covers all ten MRs, but with three seeds and no repetition of identical prompts.
- The diagnosis (ablations, attribution, replays) is offline. With Qwen, the Pooling moves and Critic routes are the model's choices, and the rates do differ (Table 6). The mechanisms are structural (count threshold, cross-ZIP borrowing, id-keyed suspect cap), and Qwen MR2 trials fail in the same ZIP-months as offline (e.g. 75202's January median at $300,500 in both). We re-ran the ablation with Qwen only for MR10 (§7); the MR2, MR4 and MR7 ablations are offline only.
- The MR10 prompt-text control has three trials (one of three violated), so it shows that noise is sufficient to cause a violation but cannot say how often.
- Seeds: 272 of 300 Qwen (MR, base input) pairs gave the same outcome under all three seeds, so the repetition sampled very little model non-determinism. Variation in the follow-up comes almost entirely from the transforms. Because the cache is keyed on the prompt, any change in the prompt text is a fresh sample, which is what MR10 exposes.
- The Qwen run was assembled from three result folders after an interrupted first attempt (MR6 had 70 of 90 trials, MR10 none). The 70 recomputed MR6 rows matched the originals exactly.
- The first Qwen attempt was killed: the job's 10 GB memory cap was exceeded and the model server was OOM-killed (`results/mr_llm_run_oom_attempt.log`). No trial had finished. The run was restarted with fewer workers. Failed LLM calls are recorded as errors, never as silent fallbacks, and replies are cached, so no result depends on the crash.
- Without the cache, a live re-run is not guaranteed byte-identical, even at temperature 0 with fixed seeds: Ollama batches 4 parallel requests, GPU arithmetic is not associative, and server versions differ.

Construct validity.

- The per-ZIP contract includes pooled listings, so a "ZIP median" can describe neighbours (0 own listings in some violating buckets). Several violations are partly a statement about this contract choice.
- Exact tolerances (10^{-9}) make any change a violation and do not grade its size. We report magnitudes in §7 to compensate.
- MR2 is partly ill-conceived (§7). Its 100% rate overstates the faults it exposes.

Statistical conclusion validity.

- The 30 base inputs come from one seeded pool and share ZIPs and months, so they are not fully independent. The Wilson interval assumes independence and is therefore optimistic.
- "Violated if any seed violates" inflates the base-input rate relative to the per-run rate. Both are reported.

External validity.

- The snapshot is synthetic (calibrated on 2021–22 Kaggle rows, with ZHVI trends). Every ZIP shares one city-level trend, and listing volumes are more volatile than a real market's.
- Dallas city only (42 ZIPs). Pooling neighbours are the numerically nearest ZIPs because the centroid file is missing from the snapshot. Real adjacency could change which buckets leak into which.
- One model (Qwen 2.5 7B, 4-bit). A stronger model may route differently.

9 Reflection: what an LLM could and could not have produced

What an LLM could have produced without running the system: the agent decomposition and loop design, the prose of the MR catalog (relations, subdomains, tolerances), and even several *predictions*. Our catalog, written before the runs, says MR2 is "partly ill-conceived on thin buckets" and that MR7 failing while MR10 holds would point to cross-ZIP pooling. That MR10 would itself fail once an LLM was in the loop was not predicted. Plausible root-cause *stories* for any violation are also cheap to generate.

What it could not have produced: anything that depends on executing this code on this snapshot. (i) The violation rates and intervals, and which three MRs top the table: MR4 at 53% was not predicted. (ii) That Pooling *alone* reproduces every violation of all three, while the jump loops are a second, hidden cause visible only with Pooling off. (iii) The Diagnoser suspect-cap mechanism. The catalog only hinted at "id-keyed exclusions". The actual cause (eight copies crowding out a `legit_rural` listing, flipping a ZIP's estimator) appeared only in a replayed trace. An AI-written first explanation of it was itself only partly right (AI_USE case 13). (iv) Operational facts: that an earlier results table was stale (old MR numbering), that a missing centroid file silently put every run on numeric ZIP order, and that the model server was OOM-killed. The pattern holds throughout: an LLM can propose hypotheses, and only execution can rank them, refute them, or reveal ones nobody proposed.

A Reproduction

Command	Produces
`make mr`	Headline Table 5 (`results/mr/summary.md`) and `docs/mr_catalog.md`; CPU only, about 15 min on 8 cores.
`make mr-llm`	Qwen table for MR2, MR4, MR7 (`results/mr_llm/`); replays from `results/llm_cache/`, otherwise needs `make ollama-server`.
`run_mrs.py --llm real --only MR1,MR3,MR5, MR6,MR8,MR9,MR10`	The remaining seven Qwen rows (`results/mr_llm_rest/`, `results/mr_llm_rest2/`); same environment variables as `make mr-llm`.
MR10 ablations	`results/mr_llm_mr10_noloops/`, `results/mr_llm_mr10_nopool/` (`jump_pct=1e9`, then also `min_listings=0`); control in `results/mr_llm_ctrl_k1e-6/`.
`make mr-diagnose`	Table 7 (`results/mr_diagnosis/`).
`python scripts/mr_replay.py MR2 0 101`	One trial with both contracts and the loops that fired.
`make app`	The Streamlit application on the frozen snapshot.